

# Aliah University

Odd (Autumn) Semester Examination 2024-25  
(For 1<sup>st</sup> year 1<sup>st</sup> Semester MCA)

Paper Name: Discrete Mathematics and Graph Theory  
Paper Code: MCAPGPC01

Full Marks: 80  
Time: 3 Hrs

## Group A (Answer all the questions)

5X2=10

1. In how many ways can you split a group of 15 persons in 4 persons and 11 persons? [CO5 BL3]
2. An undirected graph has 16 edges and all the vertices are of degree 2. Find the number of vertices. [CO6 BL3]
3. Define isomorphism between two graphs. [CO6 BL1]
4. Define self-complementary graph. [CO6 BL1]
5. Differentiate open walk and closed walk. [CO6 BL1]

## Group B (Answer any 5 questions)

5X6=30

1. a) State generalized pigeonhole principle. b) Show that in any group of 8 people at least two have birthdays which fall on the same day of the week in any given year. [CO5 BL3] [3+3]
2. a) Find the number of bit strings of length 10 that either begin with 1 or end with 0.  
b) In how many ways can six boys and four girls be arranged in straight line so that no two girls are to sit together? [CO5 BL4] [3+3]
3. Check whether the relation R defined in the set {1, 2, 3, 4, 5, 6} as  $R = \{(a, b) : b = a + 1\}$  is reflexive, symmetric or transitive. [CO1 BL4]
4. a) Explain union and intersection of graphs with examples. b) Prove that the number of vertices of odd degree in a linear graph is even. [CO6 BL3] [3+3]
5. Prove that a simple graph with n vertices and k components can have at most  $(n-k)(n-k+1)/2$  edges. [CO6 BL3]
6. Find the number of integers between 1 to 100 that are not divisible by any of the integers 2, 3, 5 or 7. [CO3 BL4]

## Group C (Answer any 4 questions)

4X10=40

1. a) There are 2500 student in a college of these 1700 have taken a course in C, 1000 have taken a course Pascal and 550 have taken course in networking further 750 have taken course in both C and Pascal, 400 have taken courses in both C and Networking and 275 have taken courses in both pascal and networking. If 200 of these students have taken course in C, pascal and Networking.  
i) How many these 2500 students have taken a course in any of these three courses C, pascal and networking?  
ii) How many of these 2500 students have not taken a courses in any of these three courses C, Pascal and networking?  
b) Use the principle of mathematical induction to show that:

$$1+3+5+\dots+(2n-1)=n^2 \quad \forall n \geq 1$$

[CO3 BL4]

[5+5]



2. a) A bowl contains 10 red balls and 10 blue balls. A woman selects balls at random without looking at them.  
 i) How many balls must she select to be sure of having at least three balls of the same color?  
 ii) How many balls must she select to be sure of having at least three blue balls?

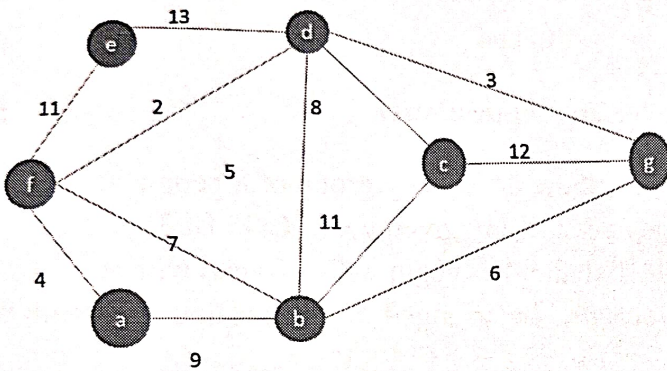
b) Solve the following IDE:

$$a_n - 2a_{n-1} + a_{n-2} = 2^n \text{ with initial conditions } a_0=2 \text{ and } a_1=1$$

[CO4 BL4]

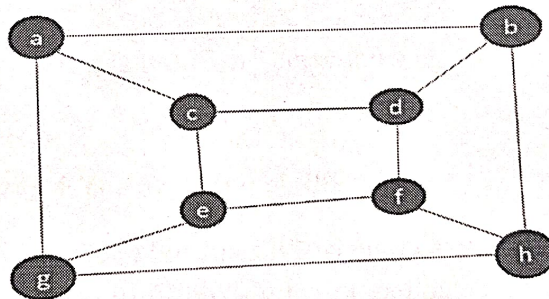
[5+5]

3. Define minimum spanning tree. Use Prim's algorithm to determine the minimal spanning tree for the following graph. What is the cost of the spanning tree? [2+7+1]



[CO6 BL4]

4. a) Define Euler graph.  
 b) Check whether the graph given in figure is Euler or not.  
 c) Determine for the graph given in figure (i) edge connectivity (ii) vertex connectivity.



[2+2+4+2] [CO6 BL4]

- d) Differentiate Hamiltonian path and Hamiltonian circuit.  
 5. a) In how many ways 3 cards can be chosen in succession from a pack of 52 cards? i) with replacement ii) without replacement  
 b) A committee is formed consisting of one representative from each of the 50 states in the United States, where the representative from a state is either the governor or one of the two senators from that state. How many ways are there to form the committee? [5+5] [CO5 BL4]



# Aliah University

## End-Semester Examination (Autumn Semester) - 2025

(For 1<sup>st</sup> Year 1<sup>st</sup> Semester MCA)

Paper Name: Database Management Systems

Full Marks: 80

Paper Code: MCAPGPC02

Time: 3 hrs

### Group-A

Answer all the following questions.

(5 × 2 = 10)

1.

- a. Explain any two types of attributes with examples. [CO1, BL1]
- b. What are the insertion anomalies in DBMS? [CO1, BL1]
- c. State the advantages of DBMS over traditional file processing systems. [CO1, BL2]
- d. Differentiate between schema and instance. [CO5, BL1]
- e. Explain the functionality of commit and rollback commands. [CO3, BL2]

### Group-B

Answer any five questions.

(5 × 6 = 30)

- 2. Different DDL commands with syntax and examples of SQL. [CO2, BL3]
- 3. Draw an E-R diagram of a University using at least five entity sets. [CO3, BL4](6)
- 4. Explain three level schema architectures of DBMS. [CO1, BL4] (6)
- 5. What is functional dependency? Find out the candidate keys and identify the normal form in the following relation R(ABCD)  $AB \rightarrow C$ ,  $AB \rightarrow D$ ,  $C \rightarrow A$ ,  $D \rightarrow B$ . [CO4, BL4] (2+4=6)
- 6. What are candidate keys? Justify BCNF is stronger than 3NF. [CO4, BL5](2+4=6)
- 7. What is Transaction? Discuss the states of the Transaction. [CO5, BL2](2+4=6)

### Group-C

Answer any four questions.

(4 × 10 = 40)

- 8. Consider the following database schema to write queries in SQL [CO2, BL5]  
(3+3+2+2=10)
  - i) Create Table STUDENT having fields Regno (Primary Key), name(text), course(text), age(integer), fees(float).
  - ii) insert five rows of data into the table STUDENT



- iii) Find the names of the students who are older than 20.
- iv) Find the regno of the student who have highest fees.

9. Discuss serializability and its type. Check whether the following schedule is Conflict serializable or not. (4+6=10) [CO5, BL4]

| T1   | T2   | T3   |
|------|------|------|
| R(X) |      |      |
|      |      | R(Y) |
| R(Y) |      |      |
|      | R(Y) |      |
|      | R(Z) |      |
|      |      | R(X) |
|      |      | W(Y) |
|      | W(Z) |      |
| R(Z) |      |      |
| W(X) |      |      |
| W(Z) |      |      |

10. Write a short note on the following. [CO3, BL2]

(i) 2PL protocol (ii) 3NF (iii) strict and cascade less recoverable schedule (3+3+4=10)

11. Explain lossless and lossy decomposition with an example. [CO4, BL2] (5+5=10)

12. What is a B tree? Build a B tree of order 4 by inserting the following data:

5, 7, 2, 10, 15, 23, 12, 19, 34, 30, 36, 40, 21, 55. (2+8= 10) [CO7, BL5]



# Aliah University

Odd (Autumn)-Semester Examination - 2024

(For 1<sup>st</sup> Semester MCA Programme)

Paper Name: Computer Organization and Architecture

Paper Code: MCAPGPC03

Full Marks: 80

Time: 3 Hrs

Group - A

(5X2=10)

1. Give a one-sentence answer -

Understand, CO1-CO4

- What component does the von Neumann machine architecture use for both instruction and data?
- What is Booth's algorithm used for?
- What are the two types of control units that can be designed in a control unit?
- Which type of memory is used to simulate a larger physical memory using secondary storage?
- What does DMA stand for?

Group - B

(Answer any 6 questions)

(6X5=30)

- Describe the von Neumann machine architecture and its significance in modern computing. **Understand, CO2**
- Describe the different types of memory based on access types. **Remember, CO3**
- Write a short note on cache memory. Discuss its role in improving system performance.
- Describe programmed I/O in brief. **Understand, CO4**
- Explain the concept and functionality of pipelining. **Remember, CO5**
- Explain memory hierarchy with suitable examples. **Understand, CO3**
- Explain the term "locality of reference" and discuss its functionality. **Understand, CO3**

Group - C

(Answer any 4 questions)

(10X4=40)

- Write short notes on
  - I/O Interface and Driver
  - Synchronous and Asynchronous Data Transfer**5+5 Understand, CO4**
- What are the different cache misses? Describe how to reduce these. **5+5 Understand, CO3**
- What is memory-mapped I/O? What do you mean by write through and write back policy? **5+5 Remember, CO4, CO3**



12. A hierarchical cache main memory subsystem has the following specifications-

- i) Cache access time of 50 nsec (ii) main memory access time of 200 nsec (iii) 75% of memory request are for read (iv) hit ratio of 0.95 for read access and the write through scheme is used.

Calculate the following: a) Average access time of the memory system considering only memory read cycle b) Average access time of the system both for read and write requests.

5+5

Apply, CO3

13. Compare hardwired control and micro-programmed control units. 5+5

Understand, CO2



# Aliah University

Odd (Autumn)-Semester Examination - 2024

(For 1<sup>st</sup> Semester MCA (CSE) Programme)

Paper Name: **Computational Mathematics**

Paper Code: MCAPGPC04

Full Marks: 80

Time: 3 Hrs

## GROUP – A

Answer all of the following

1. Choose the correct alternatives of the following questions. Answer all questions.

10 x 1 = 10

- (i) Which of the following is used to find area under curve?
- Gauss Elimination method
  - Gauss-Jordan method
  - Gauss-Seidel method
  - Trapezoidal method
- (ii) The system of equation  $x+2y=11$  and  $-2x-4y=22$  has
- Only one solution
  - Finitely many solution
  - No solution
  - Infinitely many solution
- (iii) If A is a square matrix such that  $A^2 = I$ , then  $A^{-1}$  is equal to
- 2A
  - A
  - 0
  - $A + I$
- (iv) If the points (3, -2), (x, 2), (8, 8) are collinear, then the value of x is
- 1
  - 3
  - 2
  - 5
- (v) If there is a certain number of very high values in the sample, it is preferable to compute
- Standard error
  - Standard deviation
  - Variance
  - All of the above
- (vi) A frequency distribution can be
- discrete
  - continuous
  - both a. and b.
  - None of these
- (vii) A frequency distribution having two mode is said to be
- Unimodal
  - Bimodal
  - Trimodal
  - Without mode
- (viii) If  $f(0) = 12$ ,  $f(3) = 6$  and  $f(4) = 8$ , then the interpolation function  $f(x)$  is.
- $-3x + 12$
  - $x^2 - 5x$
  - $x^3 - x^2 - 5x$
  - $x^2 - 5x + 12$
- (ix) If lower and upper limit of class are 10 and 40 respectively, then midpoint of the class is
- 25
  - 15
  - 30
  - 22.5
- (x) Mean is the measurement of
- dispersion
  - correlation
  - central values
  - None of the above

## GROUP – B

Answer any five of the following

2. Find the missing term in the following table:

5 x 5 = 25

|   |   |    |    |    |    |    |
|---|---|----|----|----|----|----|
| x | 0 | 5  | 10 | 15 | 20 | 25 |
| y | 6 | 10 | ?  | 17 | ?  | 31 |



3. Evaluate  $\int_0^1 \frac{dx}{1+x^2}$  using Simpson's  $\frac{1}{3}$  rule taking  $n = 6$ . Hence obtain the value of  $\pi$ .

4. Find the eigenvalues and corresponding eigenvectors for the matrix A. What is the dimension of the eigenspace of each eigenvalue?

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

5. Using the properties of determinants prove that

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = (a+b+c)(ab+bc+ca-a^2-b^2-c^2)$$

6. For a dataset X (3, 6, 9, 12) and Y (20, 40, 60, 80), calculate the predicted value of Y using simple linear regression when X is 8.

7. Derive the condition when regression estimator and ratio estimator are equally efficient. Define sampling error.

8. Solve the following system of equations by Gauss Elimination method:

$$3p_1 - 4p_2 + 5p_3 = 18$$

$$2p_1 - p_2 + 8p_3 = 13$$

$$5p_1 - 2p_2 + 7p_3 = 20$$

Group – C

(Answer any three of the following)

$$3 \times 15 = 45$$

9. Evaluate  $\int_1^2 \frac{dx}{1+x^2}$  taking  $h = 0.2$  using Trapezoidal rule. Can you use Simpson's rule? Justify your answer.

Find a root of  $x^4 - x - 10 = 0$  that lies between 1 and 2 using Newton Raphson method correct up to 3 decimal places. (6+2) + 7

10. Using RK method of fourth order, find  $y(0.2)$  given that  $\frac{dy}{dx} = 3e^x + 2y$ ,  $y(0) = 0$ , taking  $h = 0.1$ . Using Regula Falsi method find the root of the equation  $3x - \cos x - 1 = 0$ , correct up to three decimal places. (8+7)

11. Given that  $\frac{dy}{dx} = \log_{10}(x+y)$  with the initial condition that  $y = 1$  when  $x = 0$ . Find  $y$  for  $x = 0.2$  and  $x = 0.5$  using Euler's method. Using Finite Difference method solve the boundary value problem

$$\frac{d^2y}{dx^2} + y + 1 = 0 \text{ with } y(0) = 0, y(1) = 0. \quad (7+8)$$

12. With a suitable example explain the procedure of selecting a random sample. When to use stratified Random Sampling? Write down its advantages. 8 + (4 + 7)

13. Explain the concept of bivariate data with an example from real life. Define simple linear regression. How is the regression line determined using the least squares method? Given a dataset X (2, 4, 6) and Y (16, 32, 54), fit an appropriate curve and explain your choice. 5 + 5 + 5

14. Briefly describe the properties of determinants. Prove that  $abc = -1$ ; If  $a, b, c$  are different and

$$\begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$$

$$7 + 8$$

15. How many 5-digit telephone numbers can be constructed using the digits 0 to 9, if each number starts with 67 and no digit appears more than once? Find the number of permutations of the letters of the word ALLAHABAD. Differentiate Matrix and Determinants. Discuss laws of Matrix Algebra. 4+3+3+5



# Aliah University

Odd-Semester (Autumn) Examination - 2024

(For MCA 1<sup>st</sup> Semester)

Paper Name: Formal Languages and Automata Theory  
Paper Code: MCAPGPC05

Full Marks: 80

Time: 3 Hrs

## Group - A

1. Answer the following questions -

- a) Differentiate between DFA and NFA. (Analyze, CO1) (5X2=10)
- b) What is meant by the statement - "NPDA is more powerful than DPDA"? (Analyze, CO3)
- c) How many languages are possible for alphabet  $\Sigma = \{a, b\}$ ? (Apply, CO2)
- d) Mention all the tuples of Push Down Automata (PDA). (Remember, CO5)
- e) Mention whether the following language is Regular or Context Free, explain why? (Evaluate, CO4)

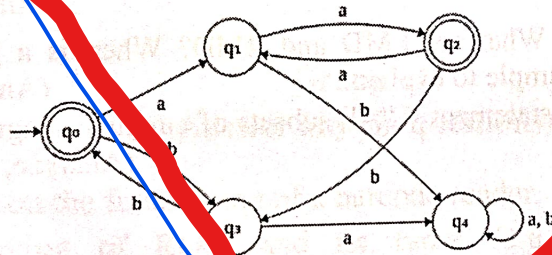
$L: \{a^n b^m c^n \mid 0 < n, m < 100000 \text{ where } \Sigma = \{a, b, c\}\}$

## Group - B

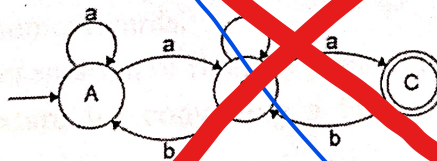
(Answer any six questions)

(6X5=30)

2. Perform state minimization of the following Finite Automata. (Apply, CO1, CO2)



3. Write Arden's theorem. Using Arden's theorem find Regular Expression for the Finite Automata given below- (Apply, CO1, CO2)



4. Construct a minimized Deterministic Finite Automata for the language given below - (Create, CO2)  
 $L: \text{All strings, having different starting and ending symbols, over alphabet } \Sigma = \{a, b\}$

5. Prove that - "Context free languages are not closed under intersection". (Analyze, CO4, CO5)

6. Construct a Turing Machine for the language below - (Create, CO4)  
 $L: \{a^n b^m \mid n \geq 1 \text{ where } \Sigma = \{a, b\}\}$

7. Write Context Free Grammar for the following language - (Apply, CO2)  
 $L: \{a^n b^m c^n \mid n, m \geq 1 \text{ where } \Sigma = \{a, b, c\}\}$

8. Using Pumping Lemma prove that - the language in **Question No-7** is not a Regular Language. (Analyze, CO4)



Group - C  
(Answer any four questions)

(4X10=40)

9. Consider the language below to answer the following questions -  
 **$L: \{a^n b^{n+m} c^m \mid n, m \geq 1, \text{ where } \Sigma = \{a, b\}\}$**
- a) Construct a Push down Automata (PDA) which accepts L. (Create, CO3, CO4)
  - b) Write a Context Free Grammar (CFG) which generates L. (Apply, CO2) 6+4
10. a) Construct a Turing Machine to accept the language given below - (Create, CO3, CO4)  
 **$L: \{ww \mid \text{where 'w' means strings formed over inputs } \Sigma = \{a, b\}\}$**
- b) Is it a deterministic or non-deterministic Turing machine? (Evaluate, CO4) 8+2
11. A regular expression is given for a language -  
 **$L: ab(a+b)^*a, \text{ where } \Sigma = \{a, b\}$**
- a) Write Regular Grammar for language L. (Apply, CO2)
  - b) Construct minimized Deterministic Finite Automata (DFA) for L. (Create, CO1) 4+6
- 12.
- a) Construct a minimized DFA for the language - (Create, CO1, CO3)  
 **$L: \{(111+1)^* \mid \Sigma = \{1\}\}$**
  - b) Construct Push Down Automata (PDA) for the language - (Create, CO3, CO4) 4+6  
 **$L: \{wcw^R \mid \text{where 'w' means strings formed over inputs } \Sigma = \{a, b\}, c \text{ is a terminal and } w^R \text{ is the reverse of } w\}$**
- 13.
- a) Define Derivation. What are LMD and RMD? When is a grammar called Ambiguous Grammar? Give example to explain. (Analyse, CO5)
  - b) Justify whether the statement - "All subsets of a regular language are Regular" is right or wrong. (Analyze, CO1, CO2, CO5) 6+4



# Aliah University

## End-Semester Examination (Autumn Semester) - 2024

(For 1<sup>st</sup> Year 1<sup>st</sup> Semester MCA Programme)

Paper Name: Foundations of Computer Science

Paper Code: MCAPGBR01

Full Marks: 80

Time: 3 hrs

### Group - A

(Answer all questions)

5 × 2 = 10

1. What are the difference between data and information? [Remembering, CO1]
2. Define volatile and non-volatile memory with examples. [Understanding, CO1]
3. Explain the difference between input devices and output devices with examples. [Understanding, CO1]
4. Convert the decimal number 25 into its binary equivalent. [Applying, CO5]
5. Define Big-O notation. [Remembering, CO4]

### Group - B

(Answer any six questions)

6 × 5 = 30

6. Write a short note on the functioning of a barcode reader. [Understanding, CO1] 5
7. Explain the working of ROM and its types (e.g., PROM, EPROM, EEPROM). [Understanding, CO1] 5
8. Discuss how cache memory improves system performance. [Analyzing, CO2] 5
9. Write a short note on memory cards. [Understanding, CO1] 5
10. Explain the binary representation of floating-point numbers. [Applying, CO5] 5
11. Write down the procedure for converting a binary number into its decimal equivalent. [Applying, CO5] 5
12. Explain the role and advantages of pointers in dynamic memory allocation with an example in C. [Applying, CO3] 5
13. Explain the differences between "call by value" and "call by address" with suitable examples in C. [Understanding, CO3] 5

### Group - C

(Answer any four questions)

4 × 10 = 40

[All parts of the same questions should be written together]

14. (a) Write down the functionality of a computer. [Understanding, CO1] 4  
(b) Discuss different categories of micro-computers. [Analyzing, CO2] 6
15. (a) Describe the various types of output devices used in multimedia systems. [Understanding, CO1] 5



(b) What are the advantages of using audio input devices like microphones?  
[Analyzing, CO2] 5

16. (a) Explain the working principle of a hard disk. [Understanding, CO1] 5  
(b) A hard disk has 10 disk plates, 1000 tracks per surface, 250 sectors per track, and 512 bytes of data can be stored per sector. Calculate the total storage capacity of the hard disk.  
[Applying, CO5] 5

17. (a) Convert the decimal number 987654 into octal and hexadecimal number systems. [Applying, CO5] 5  
(b) Multiply two binary numbers: 10101010 and 1101. [Applying, CO5] 5

18. (a) Explain the concept of recursion. What are its advantages? [Understanding, CO3] 5  
(b) Write a recursive C program to find the Greatest Common Divisor (GCD) of two numbers. [Applying, CO3] 5

19. (a) What are the advantages of using algorithms for problem-solving? [Understanding, CO1] 4  
(b) Write an algorithm to check whether a number is prime or not. [Applying, CO4] 6



# Aliah University

## Odd-Semester Examination (Autumn Semester) – 2024-25

(1<sup>st</sup> Year MCA)

Paper Name: Programming Techniques  
Paper Code: MCAPGBR02

Full Marks: 80

Time: Hrs

(General Instructions: Answers should be brief and to the point. Mention the Section name before answering it and each Section should be start from a new page and club together)

### Section - A

Answer all the questions:

(10X1=10)

| Sl. No. | Questions                                                                                                                                                                                    | Mark | CO | BT Level   |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|------------|
| 1.      | What will be output of the following program?<br><pre>int main() {     int x; x=10,20,30;     printf("%d", x);     return 0; }</pre>                                                         | 1    | 1  | Understand |
| 2.      | What is the output of the following code?<br><pre>int main() {     int sum = 10 + 4 / 2 + 4 * 2;     printf("%d", sum);     return 0; }</pre>                                                | 1    | 1  | Understand |
| 3.      | What will be output of the following program?<br><pre>int main(){     float a=0.7;     if(a&gt;0.7){         printf("C");     }     else{         printf("C++");     }     return 0; }</pre> | 1    | 1  | Understand |
| 4.      | What will be the output of the code segment?<br><pre>int main() {     int a=2,b=7,c=10;     c= a == b;     printf("%d",c);     return 0; }</pre>                                             | 1    | 1  | Understand |
| 5.      | The associability of = operator is _____.<br>a.) Right to Left b.) Left to Right c.) None of these d.) All of above                                                                          | 1    | 2  | Remember   |
| 6.      | Which operator has HIGHEST priority?<br>a.) () b.)    c.) % d.) ++                                                                                                                           | 1    | 2  | Remember   |
| 7.      | Which is a Ternary Operator? a.) ?: b.) * c.) sizeof d.) ^                                                                                                                                   | 1    | 4  | Remember   |



|     |                                                                                                                    |   |   |            |
|-----|--------------------------------------------------------------------------------------------------------------------|---|---|------------|
| 8.  | a+=4 means _____.<br>a.) a=a+4    b.) a=4    c.) a+4    d.) a=4+4                                                  | 1 | 4 | Remember   |
| 9.  | Maximum number of elements in the array declaration "int a[5][8];" is _____.<br>a.) 28    b.) 32    c.) 5    d.) 4 | 1 | 1 | Understand |
| 10. | Which is the correct way to declare a pointer?<br>a.) int ptr;    b.) *int *ptr;    c.) int *ptr;    d.) *int ptr; | 1 | 2 | Remember   |

**Section - B**  
(Answer any five questions)

(6X5=30)

| SL No. | Questions                                                                                                                                                | Marks | CO | BT Level             |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----------------------|
| 11.    | a) What is an operator? b) Explain relational operator with example.                                                                                     | 2+4   | 2  | Remember             |
| 12.    | Distinguish between the following pairs:<br>a) getchar() and scanf() functions<br>b) %s and %c specifications for reading                                | 3+3   | 2  | Remember, understand |
| 13.    | a) Describe four basic data types in C.<br>b) Define pre-increment and post increment operator in C.                                                     | 4+2   | 2  | Remember             |
| 14.    | Write a C program to read a four digit integer and print the sum of its digits.                                                                          | 4     | 1  | Apply                |
| 15.    | a) main() is a user-defined function. How does it differ from other user-defined function?<br>b) Difference between pass by reference and pass by value? | 3+3   | 3  | Understand           |
| 16.    | a) Difference between formal argument and actual argument?<br>b) & operator and * operator                                                               | 3+3   | 2  | Remember             |
| 17. ✓  | a) What is an array?<br>b) What are the different ways for initializing 1D array?                                                                        | 2+4   | 5  | Remember             |
| 18.    | a) What is a pointer?<br>b) Represent a two-dimensional array using pointer.                                                                             | 2+4   | 5  | Remember, Understand |

**Section - C**  
(Answer any four questions)

(10X4=40)

| Sl No. | Questions                                                                                                                                  | Marks     | CO  | BT Level             |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----|----------------------|
| ✓ 19.  | a) Write an Algorithm, and program code for area of a circle.<br>b) What are the properties of an Algorithm?                               | 7+3       | 1   | Apply                |
| 20.    | a) What is a function?<br>b) Explain the different elements of a function<br>c) What is prototyping? Why is it necessary?                  | 2+4+(2+2) | 6,3 | Remember, understand |
| ✓ 21.  | Compare parameter passing techniques of Call by Value and Call by reference with an example.                                               | 10        | 4   | Understand           |
| ✓ 22.  | a) Explain the differences between while and do-while loop with a suitable example.<br>b) Define the ?: operator with a suitable example.  | 6+4       | 3   | Understand, Remember |
| 23.    | a) Write a program to print odd numbers from 1 to 100 using for loop.<br>b) Explain the decision making statement in C with an example.    | 5+5       | 3   | Apply, Understand    |
| 24.    | a) What is a string? How do you declare and initialize a string?<br>b) List and explain any three string handling functions with examples. | 5+5       | 5   | Understand, Apply    |